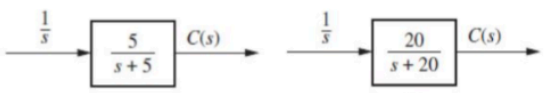
1. Find the output response, $c(t)$, for each of the systems shown below. Also find the time constant, rise time, and settling time for each case. (20 Points):



(a) (b)

2. For each of the transfer functions shown below, find the locations of the poles and zeros, plot them on the s-plane. State the nature of each response (overdamped, underdamped, and so on). (20 Points):

$$a)\ T(s) = \frac{5(s+2)}{s^2+9s+18}, \qquad b)\ T(s) = \frac{20}{s^2+6s+10}$$

$$c)\ T(s) = \frac{s+4}{s^2+9} \qquad d)\ T(s) = \frac{s}{s^2+20s+100}$$

1) $G(s) = \frac{a}{s+a}$

$R(s) = \frac{1}{s}$

$C(s) = \frac{a}{s(s+a)} = \frac{1}{s} - \frac{1}{s+a}$

$c(t) = 1 - e^{-at}$

$\tau = \frac{1}{a}$

$T_r = \frac{2.2}{a}$

$T_s = \frac{4}{a}$

a) $G(s) = \frac{5}{s+5}$

Pole: $a = 5$

Output Response: $c(t) = 1 - e^{-5t}$

$\tau = \frac{1}{5} = 0.2$ second

$T_r = \frac{2.2}{5} = 0.44$ second

$T_s = \frac{4}{5} = 0.8$ second

b) $G(s) = \frac{20}{s+20}$

$a = 20$

Output: $c(t) = 1 - e^{-20t}$

$\tau = \frac{1}{20} = 0.05$ second

$T_r = \frac{2.2}{20} = 0.11$ second

$T_s = \frac{4}{20} = 0.2$ second

2) a) $T(s) = \dfrac{5(s+2)}{s^2+9s+18}$

Zeros: set numerator to zero, $s+2=0$, $s=-2$

Poles: $s^2+9s+18 = (s+3)(s+6)$, $s=-3, -6$

S-plane plot: -2 on real axis

$-3, -6$ on real axis

$\zeta > 1$, overdamped, two distinct, real poles

b) $T(s) = \dfrac{20}{s^2+6s+10}$

Zeros: none

Poles: $s^2+6s+10=0$ $s = \dfrac{-6 \pm \sqrt{36-40}}{2} = -3 \pm j1$

S-plane: $(-3, j1)$ $(-3, -j1)$ two complex plane

$0 < \zeta < 1$, underdamped, poles complex and conjugate with negative real part

c) $T(s) = \frac{s+4}{s^2+9}$

Zeros: $s+4$, $s = -4$

Poles: $s^2+9=0$, $s = \pm j3$

s-plane: -4 on real axis, two img axis $(0, j3)$, $(0, -j3)$

Poles purely imaginary, no real part, system is undamped

$\zeta = 0$

d) $T(s) = \frac{s}{s^2+20s+100}$

$s = 0$

Poles: $s^2+20s+100 = (s-10)^2$

s-plane: double -10 on real axis

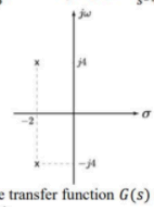
$\zeta = 1$, two identical, real poles on the negative real axis

system is critically damped

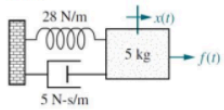
3. For each of the following transfer functions and given pole plot, find $\zeta, \omega_n, T_s, T_p$, and $\%OS$. (30 Points):

a) $T(s) = \frac{16}{s^2+3s+16}$, b) $T(s) = \frac{0.04}{s^2+0.02s+0.04}$,

c)



4. For the following system, find the transfer function $G(s) = X(s)/F(s)$ and then, find $\zeta, \omega_n, T_s, T_p$, and $\%OS$. (30 Points)



3) a) $\frac{16}{s^2+3s+16}$

$\omega_n^2 = 16$, $\omega_n = 4$

$2\zeta\omega_n = 3$, $2\zeta(4) = 3$, $\zeta = 0.375$

$\zeta\omega_n$ (real part of the pole) $= 1.5$

ω_d (imaginary part of the pole) $= \omega_n\sqrt{1-\zeta^2}$

$= 3.708$

$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{1.5} = 2.67s$

$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{3.708} = 0.847s$

$\%OS = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100$

$= 0.28059 \times 100$

$= 28.059\%$

b) $T(s) = \dfrac{0.04}{s^2 + 0.02s + 0.04}$

$\omega_n^2 = 0.04, \quad \omega_n = 0.2$

$2\zeta\omega_n \Rightarrow 2\zeta(0.2) = 0.02, \quad \zeta = 0.05$

$T_s = \dfrac{4}{\zeta\omega_n} = \dfrac{4}{(0.05)(0.2)} = 400\,s$

$T_p = \dfrac{\pi}{\omega_d} = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 15.73\,s$

$\%OS = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100$

$= 85.4\%$

c) $-2 \pm j4$

$s = -\zeta\omega_n \pm j\omega_d$

$\sigma_d = \zeta\omega_n = 2$

$\omega_d = 4$

$\omega_n = \sqrt{\sigma_d^2 + \omega_d^2}$

$= \sqrt{2^2+4^2} = 4.47\ \text{rad/s}$

$\zeta = \dfrac{\sigma_d}{\omega_n} = \dfrac{2}{\sqrt{20}} = 0.447$

$T_s = \dfrac{4}{2} = 2\,s$

$T_p = \dfrac{\pi}{4} = 0.785\,s$

$\%OS = e^{-\frac{2\pi}{4}} \times 100$

$= 20.8\%$

4) $G(s) = \frac{X(s)}{F(s)}$

$M\ddot{x}(t) + f_v \dot{x}(t) + Kx(t) = f(t)$

$M = 5,\ f_v = 5,\ K = 28$

$5\ddot{x}(t) + 5\dot{x}(t) + 28x(t) = f(t)$

$(5s^2 + 5s + 28)X(s) = F(s)$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1 \times \frac{1}{5}}{5s^2 + 5s + 28 \times \frac{1}{5}}$$

$$= \frac{0.2}{s^2 + s + 5.6}$$

$\omega_n^2 = 5.6,\ \omega_n = 2.366$

$2\zeta\omega_n = 1,\ \zeta = 0.211$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.5} = 8s$$

$$T_p = \frac{\pi}{\omega_d}$$

$$= 1.358s$$

$$\%OS = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100$$

$$= 0.5076 \times 100$$

$$= 50.76\%$$